MAT61-2

Mathematical Modeling & Decision Science

Formulation, Uncertainty, Sensitivity, and Validation

Translate ambiguous problems into explicit mathematical structures, quantify uncertainty, and test robustness before recommending action.

Albert School — 30 hours

Contents

1 Model formulation

A decision model translates a verbal business problem into an explicit mathematical structure. The formulation step produces four artifacts.

Definition 1 (Decision model) A decision model consists of:

- **Decision variables** $x = (x_1, ..., x_n)$: the quantities the decision maker controls.
- **Objective function** $f(x)$: a scalar measure of outcome to be maximized or minimized.
- **Constraints** $g_i(x) \le b_i$: the conditions a feasible decision must satisfy.
- **Assumptions**: the documented simplifications under which the model is taken to represent reality.

The three model families used in this course are:

- **Linear programs (LP)**: linear objective and linear constraints, used for static resource-allocation decisions.
- **Ordinary differential equations (ODE)**: a dynamic state evolving in continuous time, $y' = f(t, y)$.
- **Stochastic decision models**: outcomes depend on random quantities whose distributions must be specified.

Every formulation must state which variables are decisions, which are parameters (fixed inputs subject to later sensitivity analysis), and which are random. Assumptions are documented explicitly so that the validation stage (Chapter 9) can later state the conditions under which the recommendation should not be trusted.

2 Dynamic modeling with ODEs

A continuous process whose rate of change depends on its current state is modeled as an initial value problem

$$y'(t) = f(t, y(t)), \quad y(t_0) = y_0.$$

Growth, decay, and saturation are the three standard shapes:

$$y' = ky \quad \text{(growth/decay)}, \quad y' = ky\left(1 - \frac{y}{M}\right) \quad \text{(saturation)}.$$

No analytical solution is required; trajectories are obtained numerically.

Definition 2 (Euler method) Given step size h and $t_{n+1} = t_n + h$, the explicit Euler method computes

$$y_{n+1} = y_n + h\, f(t_n, y_n).$$

The Euler method is implemented in Python and the trajectory is interpreted for the decision at hand (for example, the time at which the state crosses a threshold).

```python
import numpy as np

def euler(f, t0, y0, h, n):
    t, y = t0, y0
    ts, ys = [t0], [y0]
    for _ in range(n):
        y = y + h * f(t, y)
        t = t + h
        ts.append(t); ys.append(y)
    return np.array(ts), np.array(ys)

# Example: saturating growth y' = 0.5 y (1 - y/100), y(0)=5
f = lambda t, y: 0.5 * y * (1 - y / 100)
ts, ys = euler(f, 0.0, 5.0, 0.1, 200)
```

Common pitfall The explicit Euler method accumulates error of order h per unit time and can become unstable when h is too large relative to the rate k. Reduce h and confirm the trajectory does not change materially before using it for a decision.

3 LP-based decision models

Linear programming models static decisions in which a linear objective is optimized over linear constraints. The standard form is

$$\max_{x} \quad c^\top x \quad \text{subject to} \quad Ax \le b, \quad x \ge 0.$$

Resource allocation, transportation, production planning, and staffing are all expressed in this form: x are activity levels, c are per-unit returns, and each row of $Ax \le b$ is a resource limit.

Definition 3 (Dual and shadow price) Each constraint i has an associated dual variable $y_i \ge 0$. At the optimum, y_i equals the **shadow price** of resource i: the rate of change of the optimal objective value per unit increase of the right-hand side b_i, valid within a range of b_i over which the optimal basis is unchanged.

Sensitivity interpretation. A binding constraint ($A_i x = b_i$) has, in general, a positive shadow price: relaxing it improves the objective. A non-binding constraint has shadow price zero. The shadow price is the marginal value of the constrained resource and bounds what one should pay to acquire more of it.

An assumption audit accompanies every LP: linearity of returns and costs, divisibility of x, certainty of the coefficients, and additivity of the contributions. Each assumption that fails (for example integer-only decisions, or returns that are nonlinear at scale) flags a limit on the model's validity.

4 Decision under uncertainty

Let actions be indexed by a and states of nature by s, with payoff matrix $V(a,s)$. The criteria below select an action without sequential structure.

Definition 4 (Expected value) If state s occurs with probability $p(s)$, the expected value of action a is

$$\mathrm{EV}(a) = \sum_s p(s)\, V(a,s),$$

and the EV criterion selects $a^* = \arg\max_a \mathrm{EV}(a)$.

Definition 5 (Minimax (maximin)) Without probabilities, the maximin criterion selects the action whose worst outcome is best:

$$a^* = \arg\max_a \min_s V(a,s).$$

For a loss matrix the symmetric criterion is **minimax**: minimize the worst-case loss.

Definition 6 (Minimax regret) Define regret $R(a,s) = \max_{a'} V(a',s) - V(a,s)$. The minimax-regret criterion selects

$$a^* = \arg\min_a \max_s R(a,s).$$

When to use which. Expected value is appropriate when probabilities are credible and the decision is repeated so that averages are meaningful. Maximin is appropriate under deep uncertainty or strong risk aversion, where only the worst case matters. Minimax regret targets avoidance of ex-post second-guessing relative to the best action that could have been taken.

Definition 7 (EVPI) The **expected value of perfect information** is

$$\mathrm{EVPI} = \underbrace{\sum_s p(s) \max_a V(a,s)}_{\text{EV with perfect info}} - \underbrace{\max_a \sum_s p(s) V(a,s)}_{\text{EV best action}}.$$

It is the most one should pay to learn the true state before deciding.

5 Probabilistic decision trees

A decision tree represents sequential decisions and the resolution of uncertainty over time. It alternates **decision nodes** (squares, where the decision maker chooses a branch) and **chance nodes** (circles, where nature selects a branch with given probability). Payoffs are placed at the leaves.

Definition 8 (Fold-back) The optimal policy is computed by evaluating the tree from the leaves backward to the root:

- at a chance node, take the expected value over its branches;
- at a decision node, take the maximum over its branches and record the maximizing choice.

The value at the root is the optimal expected payoff; the recorded choices form the optimal policy.

Sensitivity. After computing the optimal policy, vary one probability or one payoff at a time and recompute. The **breakeven** value of a parameter is the value at which the recommended action changes. A recommendation that flips within the plausible range of a probability or payoff is fragile and must be reported as such.

6 Simulation-based policy evaluation

When the outcome distribution of a policy has no closed form, it is estimated by Monte Carlo simulation.

Definition 9 (Monte Carlo estimate) To evaluate a policy whose outcome is $g(X)$ with X random, draw independent replications $X_1, \ldots, X_N$ and report

$$\hat{\mu} = \frac{1}{N}\sum_{i=1}^{N} g(X_i), \quad \mathrm{SE} = \frac{s}{\sqrt{N}},$$

where s is the sample standard deviation of the $g(X_i)$. An approximate 95% confidence interval is $\hat{\mu} \pm 1.96\,\mathrm{SE}$.

```
import numpy as np
rng = np.random.default_rng(0)

def evaluate(simulate_outcome, N):
    g = np.array([simulate_outcome(rng) for _ in range(N)])
    mu = g.mean()
    se = g.std(ddof=1) / np.sqrt(N)
    return mu, (mu - 1.96 * se, mu + 1.96 * se)
```

A scenario analysis specifies the set of scenarios and their generating mechanism, simulates outcomes under the policy, and reports the distribution of results — not only the mean but the spread and the confidence interval for the mean. The number of replications N controls the width of the interval, which shrinks at rate $\frac{1}{\sqrt{N}}$.

7 Continuous-time decision modeling

Irreversible decisions under continuously evolving uncertainty are framed using a continuous-time uncertainty model. The standard driver is geometric Brownian motion (GBM) from the paired Stochastic Calculus course (MAT61-4):

$$dS_t = \mu S_t\,dt + \sigma S_t\,dW_t,$$

whose scenarios S_t are simulated and fed into the decision.

Real-option framing. An irreversible decision (invest now, or wait) carries an **option value**: the right but not the obligation to act later, once more of the uncertainty has resolved. Acting immediately forgoes this option. The comparison of interest is:

- **Discrete-scenario model**: a small number of states with assigned probabilities, evaluated by expected value or a decision tree (Chapters 4–5).
- **Continuous-time model**: a GBM scenario ensemble, evaluated by simulation.

The two models are compared on the same decision: the discrete model is transparent and easy to audit but coarse; the continuous-time model captures the timing/option value of waiting at the cost of additional assumptions (μ, σ, and the GBM form itself).

8 Robustness analysis

Robustness analysis asks how the recommendation responds to perturbations of the parameters.

Definition 10 (Perturbation map) Let $V^*(\theta)$ be the optimal value as a function of parameter vector θ. The perturbation map collects the partial derivatives

$$\frac{\partial V^*}{\partial \theta_j}$$

evaluated at the baseline, ranking parameters by the sensitivity of the optimum to each.

The analysis has three components.

- **Perturbation map**: compute $\partial V^*/\partial \theta_j$ (analytically, or numerically by finite differences) for each parameter and identify the parameters to which the recommendation is most sensitive.
- **Stress testing**: evaluate the recommendation under scenario-based shocks to several parameters jointly, not only one at a time.
- **Breakeven analysis**: for each critical parameter, find the value at which the recommended action changes, and compare that value with the plausible range.

The output is the region of parameter space over which the recommendation holds, together with an explicit list of the fragile parameters — those whose plausible variation is enough to change the decision.

9 Model validation

Validation tests the model against data it was not fitted to and diagnoses model risk.

Definition 11 (Back-testing and out-of-sample error) Partition historical data into an in-sample period (used to calibrate) and an out-of-sample period (held out). **Back-testing** applies the calibrated model over the out-of-sample period and compares predictions $\hat{y}_t$ with realized outcomes y_t. The **out-of-sample prediction error** is a summary such as

$$\mathrm{RMSE} = \sqrt{\frac{1}{m}\sum_t (\hat{y}_t - y_t)^2}.$$

Calibration check. For a probabilistic model, compare predicted frequencies with observed frequencies: events assigned probability p should occur about a fraction p of the time. A systematic gap indicates miscalibration.

Model risk. Diagnostics include comparing in-sample and out-of-sample error (a large gap indicates overfitting), checking calibration, and verifying that the documented assumptions (Chapter 1) hold over the validation period. The validation memo states the conditions under which the model fails.

10 Strategic interaction (minimal core)

When the outcome depends on the choices of other rational agents, the decision is **strategic** and single-agent criteria are insufficient. This chapter is overview scope only.

Definition 12 (Dominant strategy) A strategy a_i of player i **dominates** $a_{i'}$ if it yields a payoff at least as high against every choice of the opponents, and strictly higher against at least one. A **dominant strategy** dominates all alternatives.

Definition 13 (Nash equilibrium) A profile of strategies $(a_1^*, \ldots, a_n^*)$ is a **Nash equilibrium** if no player can improve their payoff by unilaterally changing their own strategy, holding the others fixed.

A decision model requires game-theoretic reasoning when the relevant uncertainty is the deliberate choice of another agent rather than nature. The strategic structure constrains the decision: a recommendation that ignores the opponents' best responses is not credible, since the assumed state of the world will not persist once the other agents react.

11 Capstone dossier

The course deliverable is a dossier with four mandatory outputs, each corresponding to a stage of the preceding chapters.

1. **Formal model specification** — decision variables, objective, constraints, and documented assumptions (Chapter 1), in the appropriate family (LP, ODE, or stochastic).
2. **Uncertainty model with calibration assumptions** — the distributions or scenarios used (Chapters 4–7) and how their parameters were calibrated.
3. **Sensitivity / robustness analysis** — perturbation map, stress tests, and breakeven values; the region in which the recommendation holds and the fragile parameters (Chapter 8).
4. **Validation memo** — back-testing and out-of-sample results, calibration checks, and an explicit statement of the conditions under which the recommendation should not be trusted (Chapter 9).